

Soundness of the N -Hop Abstraction

A refinement theorem transferring routing safety to the concrete protocol for arbitrary path length

This section upgrades the central claim from “we verified an idealized N -hop model” to a *theorem*: safety proved on the abstract, unbounded-length routing model *transfers* to the concrete, full-cryptography protocol, for every path length N . The argument decomposes the problem into a part that is cryptographic but *local* (discharged in the concrete model at bounded size) and a part that is unbounded but *structural* (discharged in the abstract model for all N), and composes them by a pen-and-paper refinement.

Two models over a shared action alphabet

Definition 1 (Concrete model $\mathcal{C}$). *The full-cryptography, fixed-position theory multihop.spthy : HTLC offers are signed messages sent over an Out/In network under a Dolev–Yao adversary with key compromise (Compromise_Ltk). Forwarding is authenticated by verifying the previous hop’s signature.*

Definition 2 (Abstract model $\mathcal{A}$). *The linear-fact routing theory ($\text{multihop_nhop_fees.spthy}$): the HTLC is handed hop-to-hop as an internal linear fact $\text{Route}(\text{prev}, \text{me}, \text{ptr}, y)$ (an idealised channel) rather than a signed broadcast. A single generic forward rule fires at every hop, so paths have arbitrary length N .*

Both theories emit the same safety-relevant action facts $\Sigma = \{\text{ForwardHTLC}, \text{Redeemed}, \text{Refunded}\}$ with identical arities; the intermediary-safety properties are stated over Σ only. (The concrete model additionally emits the position-indexed HTLCForwarded1/2 ; the projection below collapses both to the abstract generic HTLCForwarded , but no safety lemma depends on the index.)

The trap, and the local authentication lemma

Remark 1 (Why abstraction is not automatically sound). *Internalising the network as a trusted Route fact removes Dolev–Yao power: $\mathcal{A}$ has strictly fewer behaviours than $\mathcal{C}$. Safety on $\mathcal{A}$ therefore does not, by itself, imply safety on $\mathcal{C}$. The gap is exactly the adversary’s ability to forge or replay a forwarded HTLC; closing it requires a cryptographic authentication guarantee in $\mathcal{C}$.*

Lemma 1 (AUTH, per-hop authentication; machine-checked in $\mathcal{C}$). *For every forwarding step, the incoming HTLC accepted by a node traces back to a genuine offer/forward from the adjacent previous hop, unless that hop’s long-term key is compromised. Formally (one hop of the path), for the first and second forwarding positions:*

$$\begin{aligned} \text{HTLCForwarded1}(F_1, F_2, S, p, y, v_{in}, v_{out}) &\Rightarrow (\exists p', t. \text{HTLCOffered}(S, F_1, p', y, v_{in})@t) \vee \text{LtkCompromised}(S), \\ \text{HTLCForwarded2}(F_2, R, F_1, p, y, v_{in}, v_{out}) &\Rightarrow (\exists S, p', v. \text{HTLCForwarded1}(F_1, F_2, S, p', y, v, v_{in})@t) \vee \text{LtkCompromised}(S), \end{aligned}$$

Status. Machine-checked in `multihop.spthy` as `Forward1_Requires_Offer` (266 steps) and `Forward2_Requires_Forward1` (274 steps), together with their compromise-free variants `*_Honest`. Each quantifies over a single adjacent pair ($prev, me$) and never over the path or N ; the forward rules verify only the previous hop’s signature and read only the two adjacent channels. Hence `AUTH` is *local*. $\square$

Remark 2 (Locality is what makes `AUTH` tractable). *A generic single signed-forward rule that fires at every hop makes the ‘`htlc`’ term self-referential (a forwarded offer is the output of forwarding another offer). Tamarin’s origin analysis then fails to terminate at the sources stage: we verified that on such a rule no lemma completes—not `AUTH`, not its honest variant, not even a two-hop reachability witness—at 300 s, and a hand-written [`sources`] lemma does not rescue it (420 s). `AUTH` is provable precisely because, taken per hop, it is bounded: it looks back one adjacent step and does not unfold the path. This is the crux of the whole argument.*

Projection and the refinement theorem

Definition 3 (Projection π). π maps a concrete trace to an abstract one by (i) keeping every Σ -action verbatim and (ii) deleting all network/crypto events (`Out`, `In`, `KU`, signature checks). Position-indexed `HTLCForwarded1/2` are relabelled to the generic `HTLCForwarded`.

Lemma 2 (Per-hop simulation). *Under `AUTH`, every concrete forwarding step has a matching abstract step: an honestly-authenticated incoming HTLC (Lemma 1) is exactly what licenses the abstract `Route` hand-off, so π of a concrete step is a legal abstract step. The adversarial case (a forged/replayed HTLC) is ruled out by `AUTH` except under key compromise, which both models expose identically via `LtkCompromised`.*

Theorem 1 (Soundness of the abstraction). *Let φ be a safety property over Σ proved on $\mathcal{A}$ for all path lengths N . Then φ holds on $\mathcal{C}$: for every concrete trace t , $\pi(t)$ is an abstract trace (Lemmas 1, 2, by induction over the forwarding steps of t), so if $\pi(t)$ satisfies φ then so does t . Equivalently, every concrete violation projects to an abstract violation; contrapositively, abstract safety transfers. Hence the intermediary-safety and atomicity properties, verified structurally on $\mathcal{A}$ for arbitrary N , hold for the concrete signed-message protocol at arbitrary N .*

Structure. By induction on the number of forwarding steps in t . Base case: zero forwards, $\pi(t)$ trivially abstract. Step: extend t by one concrete forward; by Lemma 2 (using Lemma 1 to discharge the forgery case) its projection extends $\pi(t)$ by one legal abstract forward. The Σ -actions are preserved verbatim, so any Σ -violation in t appears in $\pi(t)$. $\square$

What is machine-checked, and what is pen-and-paper

Ingredient	Where	Status
<code>AUTH</code> , per-hop, local	$\mathcal{C}$ (<code>multihop.spthy</code>)	machine-checked (266 / 274 steps)
All- N routing safety (φ)	$\mathcal{A}$ (<code>*_nhop_fees</code>)	machine-checked (28/29; 30/30; 551-step <code>Intermediary_Never_</code>
Σ -alignment / projection π	both	by inspection (Σ shared verbatim)
Simulation + induction (Thm. 1)	meta	pen-and-paper
Generic single-rule <code>AUTH</code>	—	<i>boundary</i> : sources non-termination (Rem. 2)

The meta-theorem is pen-and-paper by necessity: Tamarin cannot quantify over all N or relate two theories (the same reason key-management meta-theorems are argued externally). Its two

hypotheses, however, are both machine-checked, and its soundness rests on the *locality* of AUTH (Remark 2)—which must be, and has been, confirmed by inspection of every forward rule: each references only its adjacent pair. If any forward rule depended on global path structure the argument would break; none does.